Lesson notes · for you

Pass it on: mesh networks

IoT · Level 4 · Lesson 2

Time
45 min

Materials
Paper, a pencil and a ruler; A piece of string and scissors; Optional: friends, to try a human mesh in the playground

Learning goals

- Explain how a mesh passes messages hop by hop to a gateway.
- Find the weak points in a network and fix them with relays.
- Say why a mesh keeps working when one device fails.

Activity: Paper mesh

1. Draw a place you know, such as your school or a farm. Mark ten sensors as dots and the gateway as a square. Choose a scale, such as 1 cm for 20 m.
2. At that scale, a 200 m radio range is 10 cm of string. Cut it carefully, away from other people.
3. Hold one end on each sensor in turn and draw a line to every point the other end reaches. Each line is a radio link.
4. Circle any sensor with no line (cut off) and any with only one line (one fallen neighbour would cut it off).
5. Add the fewest relays that give every sensor a route, and two routes where you can. Write each sensor's hop count beside it.

Use scissors carefully. If you try a human mesh outside, stay inside the school grounds and away from roads, and pass notes only to someone within arm's reach.

Discussion

- What happened to your network when you removed one sensor?
- Why do sensors near the gateway run down first?
- Where in your town would a mesh help, and what could break it?

Check yourself

- Why does a mesh keep working when one sensor fails?

Try it this week
Paper mesh: draw ten sensors and a gateway on a map, link every pair that a string as long as the radio's range can reach, find a sensor that's cut off, and add a relay to rescue it.

Worksheet

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Name _____ Class _____ Date _____

1 **Zainab's sensor by the river has only one neighbour, and floods keep knocking that neighbour over. What is the best fix?**

- ◯ Add a relay on higher ground
- ◯ Give her sensor a bigger battery
- ◯ Make her sensor send more often

2 **In a mesh, messages can route round a broken sensor, but if the gateway fails, no message gets out at all.**

- ◯ True
- ◯ False

3 **Match each weak point to its fix.**

A gap wider than the radio's range	• • • •	Give the busiest nodes bigger batteries
A sensor with only one neighbour	• • • •	Give it a solar panel and a battery
The gateway loses power at night	• • • •	Add a relay as a second neighbour
Sensors next to the gateway run flat first	• • • •	Put a relay in the gap

4 **In a farm mesh, every sensor sends one reading an hour. Predict: whose batteries run down first?**

- ◯ The sensors farthest from the gateway
- ◯ The sensors next to the gateway
- ◯ All of them at the same time

5 **Count the hops along a line of sensors.**

Sensors stand every 250 m along a road, and each radio reaches 300 m. A message from a sensor 1,000 m from the gateway needs _____ hops. If each hop adds 0.1 s, it arrives after about _____ s.

Word bank: 4 · 0.4 · 3 · 1

✎ **Where near you could a mesh help: a farm, a river, a market, a school compound? Where would you put the gateway, and what might knock a sensor over?**